This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scripter Modules, refer to the "Guide to Using Scripter Modules" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Barco
Firmware Version: N/A
Model(s): F90-4K13, F90-W13, F80-Q9

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scripter Version
2.06.0002-b002	1.4.2

Version History

Module Version	Date	Notes
1_1_0_0	1/31/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
  Example: `InterfaceName.Unidirectional = 'True'`

- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
  Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='F90-4K13')
```

SerialOverEthernetClass

```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='F90-4K13')
```

EthernetClass

```
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 9090, Model='F90-4K13')
```

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	Value
Focus	'In'	'Out'	'Stop'
Qualifier Key **'Steps'**	Qualifier Value 1 to 500 in steps of 1		

```
# Focus example
InterfaceName.Set('Focus', 'In', {'Steps': 1})
```

Command	Value
Illumination	0 to 100 in steps of 1

```
# Illumination example
InterfaceName.Set('Illumination', 100)
```

Command	Value	Value	Value
Input	'DVI 1'	'DVI 2'	'DisplayPort 1'
	'DisplayPort 2'	'HDMI'	'HDBaseT'
	'SDI'		

```
# Input example
InterfaceName.Set('Input', 'DVI 1')
```

Command	Value	Value	Value
LensShift	'Up'	'Down'	'Left'
	'Right'	'Stop'	
Qualifier Key **'Steps'**	Qualifier Value 1 to 500 in steps of 1		

```
# LensShift example
InterfaceName.Set('LensShift', 'Up', {'Steps': 1})
```

Command	Value	Value
Power	'On'	'Off'

```
# Power example
InterfaceName.Set('Power', 'On')
```

Command	Value
Shutter	None

```
# Shutter example
InterfaceName.Set('Shutter', None)
```

Command	Value	Value
TestPattern	'On'	'Off'

```
# TestPattern example
```

```
InterfaceName.Set('TestPattern', 'On')
```

Command	Value	Value	Value
TestPatternSelect	'Focus Black'	'Ansi Lumen'	'Scenergix'
	'Checker Board'	'Focus'	'Cross Hatch'
	'Monoscope'	'Outline'	'Color Bars'
	'Aspect'		

```
# TestPatternSelect example
InterfaceName.Set('TestPatternSelect', 'Focus Black')
```

Command	Value	Value	Value
Zoom	'In'	'Out'	'Stop'

Qualifier Key	Qualifier Value
'Steps'	1 to 500 in steps of 1

```
# Zoom example
InterfaceName.Set('Zoom', 'In', {'Steps': 1})
```

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
```

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
```

FeedbackHandler will be called when any qualifier gets a new status.

Command	Value	Value
ConnectionStatus	'Connected'	'Disconnected'

```
# ConnectionStatus examples
Value = InterfaceName.ReadStatus('ConnectionStatus')
InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)
```

Command	Value
Illumination	0 to 100 in steps of 1

```
# Illumination examples
InterfaceName.Update('Illumination')
Value = InterfaceName.ReadStatus('Illumination')
InterfaceName.SubscribeStatus('Illumination', None, FeedbackHandler)
```

Command	Value	Value	Value
Input	'DVI 1'	'DVI 2'	'DisplayPort 1'
	'DisplayPort 2'	'HDMI'	'HDBaseT'
	'SDI'		

```
# Input examples
InterfaceName.Update('Input')
Value = InterfaceName.ReadStatus('Input')
InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)
```

Command	Value	Value
Power	'On'	'Off'

```
# Power examples
InterfaceName.Update('Power')
Value = InterfaceName.ReadStatus('Power')
InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)
```

Command	Value	Value
TestPattern	'On'	'Off'

```
# TestPattern examples
InterfaceName.Update('TestPattern')
```

```
Value = InterfaceName.ReadStatus('TestPattern')
InterfaceName.SubscribeStatus('TestPattern', None, FeedbackHandler)
```

Command	Value	Value	Value
TestPatternSelect	'Focus Black'	'Ansi Lumen'	'Scenergix'
	'Checker Board'	'Focus'	'Cross Hatch'
	'Monoscope'	'Outline'	'Color Bars'
	'Aspect'		

```
# TestPatternSelect examples
InterfaceName.Update('TestPatternSelect')
Value = InterfaceName.ReadStatus('TestPatternSelect')
InterfaceName.SubscribeStatus('TestPatternSelect', None, FeedbackHandler)
```

Cable and Adapter Requirements

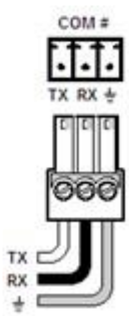
Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

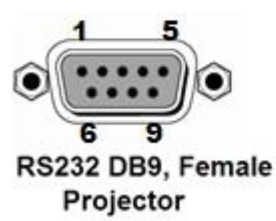
Serial communication

Port Type: RS-232
Baud Rate: 19200
Data Bits: 8
Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	⟶	2	RxD
RxD	⟵	3	TxD
GND	—	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	9090
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Undetermined

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Focus In Steps 1	{"jsonrpc": "2.0", "method": "optics.focus.stepforward", "params": {"steps": 1}, "id": 470}\x0D\x0A
Focus Out Steps 1	{"jsonrpc": "2.0", "method": "optics.focus.stepreverse", "params": {"steps": 1}, "id": 470}\x0D\x0A
Focus Stop Steps 1	{"jsonrpc": "2.0","method": "optics.focus.stop","params": {},"id": 470}\x0D\x0A
Illumination 0	{"jsonrpc": "2.0","method": "property.set","id": 300,"params": {"property": "illumination.sources.laser.power","value": 0}}\x0D\x0A
Illumination 100	{"jsonrpc": "2.0","method": "property.set","id": 300,"params": {"property": "illumination.sources.laser.power","value": 100}}\x0D\x0A
Input DVI 1	{"jsonrpc": "2.0","method": "property.set","params": {"property": "image.window.main.source", "value": "DVI 1"},"id": 423}\x0D\x0A
Input DVI 2	{"jsonrpc": "2.0","method": "property.set","params": {"property": "image.window.main.source", "value": "DVI 2"},"id": 423}\x0D\x0A
Input DisplayPort 1	{"jsonrpc": "2.0","method": "property.set","params": {"property": "image.window.main.source", "value": "DisplayPort 1"},"id": 423}\x0D\x0A
Input DisplayPort 2	{"jsonrpc": "2.0","method": "property.set","params": {"property": "image.window.main.source", "value": "DisplayPort 2"},"id": 423}\x0D\x0A
Input HDBaseT	{"jsonrpc": "2.0","method": "property.set","params": {"property": "image.window.main.source", "value": "HDBaseT"},"id": 423}\x0D\x0A
Input HDMI	{"jsonrpc": "2.0","method": "property.set","params": {"property": "image.window.main.source", "value": "HDMI"},"id": 423}\x0D\x0A
Input SDI	{"jsonrpc": "2.0","method": "property.set","params": {"property": "image.window.main.source", "value": "SDI"},"id": 423}\x0D\x0A
Lens Shift Down Steps	{"jsonrpc": "2.0", "method": "optics.lensshift.vertical.stepforward", "params": {"steps": 1}, "id": 2356}\x0D\x0A

Lens Shift Left Steps 1	{"jsonrpc": "2.0", "method": "optics.lensshift.horizontal.stepreverse", "params": {"steps": 1}, "id": 2356}\x0D\x0A
Lens Shift Right Steps 1	{"jsonrpc": "2.0", "method": "optics.lensshift.horizontal.stepforward", "params": {"steps": 1}, "id": 2356}\x0D\x0A
Lens Shift Stop Steps 1	{"jsonrpc": "2.0","method": "optics.lensshift.stop","params": {},"id": 2356}\x0D\x0A
Lens Shift Up Steps 1	{"jsonrpc": "2.0", "method": "optics.lensshift.vertical.stepreverse", "params": {"steps": 1}, "id": 2356}\x0D\x0A
Power Off	{"jsonrpc": "2.0","method": "system.poweroff","params": {},"id": 130}\x0D\x0A
Power On	{"jsonrpc": "2.0","method": "system.poweron","params": {},"id": 130}\x0D\x0A
Shutter None	{"jsonrpc": "2.0","method": "optics.shutter.toggle","params": {},"id": 310}\x0D\x0A
Test Pattern Off	{"jsonrpc": "2.0","method": "property.set","id": 141,"params": {"property": "image.testpattern.show", "value": false}}\x0D\x0A
Test Pattern On	{"jsonrpc": "2.0","method": "property.set","id": 141,"params": {"property": "image.testpattern.show", "value": true}}\x0D\x0A
Test Pattern Select Ansi Lumen	{"jsonrpc": "2.0","method": "property.set","id": 147,"params": {"property": "image.testpattern.selected", "value": "internal:Ansi Lumen"}}\x0D\x0A
Test Pattern Select Aspect	{"jsonrpc": "2.0","method": "property.set","id": 147,"params": {"property": "image.testpattern.selected", "value": "internal:Aspect"}}\x0D\x0A
Test Pattern Select Checker Board	{"jsonrpc": "2.0","method": "property.set","id": 147,"params": {"property": "image.testpattern.selected", "value": "internal:Checker Board"}}\x0D\x0A
Test Pattern Select Color Bars	{"jsonrpc": "2.0","method": "property.set","id": 147,"params": {"property": "image.testpattern.selected", "value": "internal:Color Bars"}}\x0D\x0A
Test Pattern Select Cross Hatch	{"jsonrpc": "2.0","method": "property.set","id": 147,"params": {"property": "image.testpattern.selected", "value": "internal:Cross Hatch"}}\x0D\x0A
Test Pattern Select Focus	{"jsonrpc": "2.0","method": "property.set","id": 147,"params": {"property": "image.testpattern.selected", "value": "internal:Focus"}}\x0D\x0A
Test Pattern Select Focus Black	{"jsonrpc": "2.0","method": "property.set","id": 147,"params": {"property":

	"image.testpattern.selected", "value": "internal:Focus Black"}}\x0D\x0A
Test Pattern Select Monoscope	{"jsonrpc": "2.0","method": "property.set","id": 147,"params": {"property": "image.testpattern.selected", "value": "internal:Monoscope"}}\x0D\x0A
Test Pattern Select Outline	{"jsonrpc": "2.0","method": "property.set","id": 147,"params": {"property": "image.testpattern.selected", "value": "internal:Outline"}}\x0D\x0A
Test Pattern Select Scenergix	{"jsonrpc": "2.0","method": "property.set","id": 147,"params": {"property": "image.testpattern.selected", "value": "internal:Scenergix"}}\x0D\x0A
Zoom In Steps 1	{"jsonrpc": "2.0", "method": "optics.zoom.stepforward", "params": {"steps": 1}, "id": 450}
Zoom Out Steps 1	{"jsonrpc": "2.0", "method": "optics.zoom.stepreverse", "params": {"steps": 1}, "id": 450}
Zoom Stop Steps 1	{"jsonrpc": "2.0","method": "optics.zoom.stop","params": {},"id": 450}\x0D\x0A

Appendix B. Update Commands

Illumination	{"jsonrpc": "2.0","method": "property.get","params": {"property": "illumination.sources.laser.power"},"id": 300}\x0D\x0A
Input	{"jsonrpc": "2.0","method": "property.get","params": {"property": "image.window.main.source"},"id": 423}\x0D\x0A
Power	{"jsonrpc": "2.0","method": "property.get","params": {"property": "system.state"},"id": 130}\x0D\x0A
Test Pattern Select	{"jsonrpc": "2.0","method": "property.get","params": {"property": "image.testpattern.selected"},"id": 147}\x0D\x0A